

Sparse Variable Selection in the Cox Model

Zhiyuan Zhang and Imani Elenor Beckett

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Two luminaries behind the paper



Sir David Cox

Cox PH model

Box–Cox transformation

Partial likelihood



Robert Tibshirani

LASSO

GAM

glmnet

Cox gives the survival-regression framework; Tibshirani brings shrinkage and selection into that framework.

The problem: variable selection in Cox regression

Survival studies often have

- censored survival outcomes;
- multiple clinical predictors;
- a need for prediction *and* interpretation.

Classical question

Which covariates should enter the Cox model?

Selection without shrinkage can be unstable.

many predictors  sparse prognostic Cox model

Why stepwise selection is the target

Stepwise Cox selection

- gives a sparse model;
- searches discretely over models;
- can make selected effects and Z -scores look optimistic;
- selects variables, but does not shrink them.

Tibshirani's alternative

sparse model

plus

coefficient shrinkage

through one continuous constraint path.

Lasso gives sparsity through shrinkage.

Cox partial likelihood is the object to constrain

$$\lambda(t \mid \mathbf{Z}) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{Z})$$

$$L(\boldsymbol{\beta}) = \prod_{i: \delta_i=1} \frac{\exp(\boldsymbol{\beta}^\top \mathbf{Z}_i)}{\sum_{j \in R_i} \exp(\boldsymbol{\beta}^\top \mathbf{Z}_j)}$$

Why this matters

- the baseline hazard cancels;
- estimation focuses on $\boldsymbol{\beta}$.

Tibshirani's move

- keep the same partial likelihood;
- change the estimation problem.

The move: add an L_1 constraint to partial likelihood

$$\hat{\beta}(s) = \arg \min_{\beta} \{-\ell_{\text{partial}}(\beta)\} \quad \text{s.t.} \quad \sum_j |\beta_j| \leq s$$

Large s

ordinary Cox
partial-likelihood
estimate.

Small s

stronger shrinkage and
more zero coefficients.

Standardize

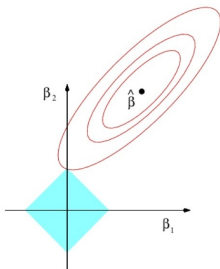
covariates before
penalization because the
constraint is
scale-sensitive.

shrinkage + exact zeros = variable selection

Why L_1 gives exact zeros

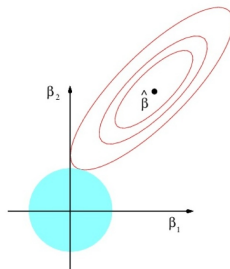
Lasso / L_1

$$\sum_j |\beta_j| \leq s$$



Ridge / L_2

$$\sum_j \beta_j^2 \leq t$$



Computation: Newton step becomes constrained WLS

Ordinary Cox fitting uses a local quadratic / Newton step.

$$\min_{\beta} (\mathbf{z} - \mathbf{X}\beta)^\top \mathbf{A} (\mathbf{z} - \mathbf{X}\beta) \quad \text{s.t.} \quad \sum_j |\beta_j| \leq s$$

ordinary Cox
Newton/IRLS step $\longrightarrow$ L_1 -constrained
WLS step

The algorithm repeatedly solves a local quadratic approximation with an L_1 constraint.

The tuning parameter controls shrinkage

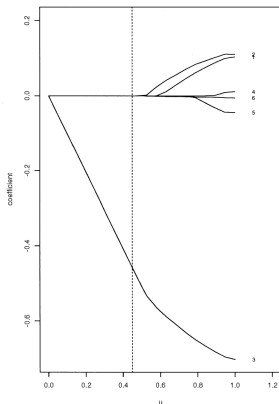
$$u = \frac{s}{\|\hat{\beta}^0\|_1} = \frac{s}{\sum_j |\hat{\beta}_j^0|}$$

where $\hat{\beta}^0$ is the ordinary Cox partial-likelihood estimate.

GCV selects a point on this path



Example 1: shrinkage path is visible



VA lung cancer data

- GCV-selected $u \approx 0.45$.
- Lasso retains only Karnofsky score.
- Stepwise also selects Karnofsky, but with a larger unshrunk coefficient.

$$\hat{\beta}_{\text{lasso}} \approx -0.47, \quad \hat{\beta}_{\text{stepwise}} \approx -0.67$$

Figure 1. VA lung cancer coefficient path.

Example 2: similar variables, more shrinkage

Mayo PBC / liver data

- 276 complete observations;
- 17 candidate predictors;
- standardized covariates;
- GCV selected $u = 0.56$.

What the example shows

- Stepwise selected several clinical predictors.
- Lasso retained a similar variable structure.
- Most effects were shrunk toward zero.

**variable structure can be similar;
effect sizes become more
conservative.**

Choosing s : approximate GCV balances fit and complexity

ASSO METHOD FOR VARIABLE SELECTION IN THE COX MODEL

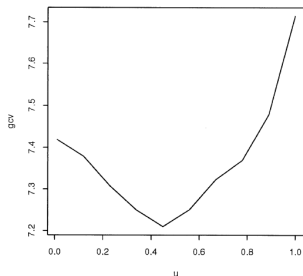


Figure 2. Approximate GCV curve used to choose the shrinkage level.

Bias–variance tradeoff

- small s : high bias, low variance;
- large s : low bias, high variance.

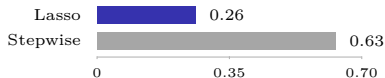
Tibshirani's solution

- approximate lasso complexity by effective degrees of freedom;
- inflate negative log partial likelihood by complexity;
- choose s by approximate GCV.

Simulation: lasso reduces variability relative to stepwise

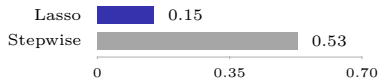
Figure 3. Median prediction MSE in Tibshirani's simulation experiments; lower is better.

Few large effects



Lasso finds sparsity and lowers prediction error.

Many small effects



Lasso wins mainly by shrinkage, not by discovering true zeros.

Shrinkage trades some bias for variance reduction.

Standard errors: useful approximation, delicate inference

What the paper does

- linearizes the lasso estimate;
- uses the IRLS approximation;
- obtains approximate standard errors.

What to be careful about

- shrinkage introduces bias;
- zeros create non-smooth behavior;
- normal intervals can be unreliable near zero.

Approximate SEs are most convincing for clearly nonzero effects.

Tibshirani's discussion: lasso is a model-building tool

The lasso helps with selection + shrinkage, but it does not replace:

Model diagnostics

linearity and PH assumptions still need checking.

Penalty design

standardization, categorical scaling, and unpenalized covariates matter.

Extensions

Bayesian/bootstrap alternatives and time-dependent covariates remain natural directions.

Cox-LASSO is a smoother model-building tool, not a replacement for survival diagnostics.

Final message: the contribution is the reframing

Tibshirani did not propose a new survival model.

Cox partial likelihood + L_1 constraint
 $\Rightarrow$ shrinkage + variable selection

1. Stepwise selection is sparse but unstable.
2. Lasso creates sparsity through shrinkage.
3. Cox-LASSO is useful, but diagnostics and inference still matter.

The paper turns Cox variable selection from model search into constrained estimation.

Optional: from 1997 limitations to later research

The original paper opened several later research directions.

1997 issue	Later response
Approximate GCV for choosing s	K-fold CV, C-index / partial-likelihood CV, and external validation
IRLS + constrained WLS can be computationally heavy	Coordinate descent and regularization paths, e.g. <code>glmnet</code>
Correlated or grouped predictors can make selection unstable	Elastic net, group lasso, and sparse-group lasso
Shrinkage bias and post-selection uncertainty	SCAD / MCP / adaptive lasso; selective inference, debiased lasso, and stability selection

The 1997 paper did not close the problem; it opened a research program in penalized survival modeling.